

5.12 Homework: Integration Review (no calculator) — Markscheme

1. Consider an arithmetic sequence with $u_1 = 0.6$ and $u_4 = 0.15$.

(a) Find the common difference, d .

$$u_1 + 3d = u_4 \quad (\text{M1})$$

$$0.6 + 3d = 0.15$$

$$d = -0.15 \quad (\text{A1})$$

(b) Find the value of k .

$$u_2 = 0.45, u_3 = 0.3 \quad (\text{A1})$$

Sum of probabilities equals 1: (M1)

$$\frac{0.6}{k} + \frac{0.45}{k} + \frac{0.3}{k} + \frac{0.15}{k} = 1 \quad (\text{A1})$$

$$\frac{1.5}{k} = 1$$

$$k = 1.5 \quad (\text{A1})$$

[6 marks]

2. Let a be a constant, where $a > 1$.

(a) Show that $a^2 + \left(\frac{a^2 - 1}{2}\right)^2 = \left(\frac{a^2 + 1}{2}\right)^2$.

Expanding the LHS: (M1)

$$a^2 + \frac{a^4 - 2a^2 + 1}{4} = \frac{4a^2 + a^4 - 2a^2 + 1}{4} \quad (\text{A1})$$

$$= \frac{a^4 + 2a^2 + 1}{4} = \left(\frac{a^2 + 1}{2}\right)^2 \quad (\text{A1})$$

(b) Find an expression for the area of the triangle in terms of a .

The right angle is opposite the hypotenuse $\frac{a^2 + 1}{2}$, so the legs are a and $\frac{a^2 - 1}{2}$. (M1)

$$\text{Area} = \frac{1}{2} \cdot a \cdot \frac{a^2 - 1}{2} = \frac{a^3 - a}{4} = \frac{a(a^2 - 1)}{4} \quad (\text{A1})$$

[5 marks]

3. The function f is defined by $f(x) = \frac{7x + 7}{2x - 4}$ for $x \in \mathbb{R}, x \neq 2$.

(a) Recognizing $f(x) = 0$: (M1)

$$7x + 7 = 0 \implies x = -1 \quad (\text{A1})$$

(b)(i) Vertical asymptote: $x = 2$ (A1)

(b)(ii) Horizontal asymptote: $y = \frac{7}{2}$ (A1)

(c) Interchanging x and y : (M1)

$$x = \frac{7y + 7}{2y - 4}$$

$$2xy - 4x = 7y + 7$$

$$2xy - 7y = 4x + 7 \quad (\text{A1})$$

$$y(2x - 7) = 4x + 7$$

$$f^{-1}(x) = \frac{4x + 7}{2x - 7} \quad (\text{A1})$$

[7 marks]

4. (a) Show that $\cos 2x = \sin x$ can be written as $2\sin^2 x + \sin x - 1 = 0$.

Using $\cos 2x = 1 - 2\sin^2 x$:

$$1 - 2\sin^2 x = \sin x$$

$$2\sin^2 x + \sin x - 1 = 0 \quad (\text{A1})$$

(b) Solve $\cos 2x = \sin x$ for $-\pi \leq x \leq \pi$.

$$\text{Factoring: } (2\sin x - 1)(\sin x + 1) = 0 \quad (\text{M1})$$

$$\sin x = \frac{1}{2} \text{ or } \sin x = -1 \quad (\text{M1})$$

$$\text{From } \sin x = -1: x = -\frac{\pi}{2} \quad (\text{A1})$$

$$\text{From } \sin x = \frac{1}{2}: x = \frac{\pi}{6}, x = \frac{5\pi}{6} \quad (\text{A1})$$

$$x = -\frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6} \quad (\text{A1})$$

[6 marks]

5. Find the least positive value of x for which $\cos\left(\frac{x}{2} + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$.

Let $\theta = \frac{x}{2} + \frac{\pi}{3}$.

Reference angle for $\cos \theta = \frac{1}{\sqrt{2}}$: $\frac{\pi}{4}$ (A1)

$\theta = \frac{\pi}{4}$ gives $\frac{x}{2} = \frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$, so $x = -\frac{\pi}{6}$ (negative, rejected) (M1)(R1)

$\theta = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$ (A1)

$\frac{x}{2} = \frac{7\pi}{4} - \frac{\pi}{3} = \frac{21\pi - 4\pi}{12} = \frac{17\pi}{12}$

$x = \frac{17\pi}{6}$ (A1)

[5 marks]

6. Events A and B are such that $P(A) = 0.3$ and $P(B) = 0.8$.

(a) If independent: $P(A \cap B) = 0.3 \times 0.8 = 0.24$ (A1)

(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1.1 - P(A \cap B)$ (A1)

Since $P(A \cup B) \leq 1$: (M1)

$1.1 - P(A \cap B) \leq 1$

Minimum value of $P(A \cap B) = 0.1$ (A1)

(c) A is a subset of B (since $P(A) < P(B)$), so $P(A \cap B) = P(A)$. (R1)

Maximum value of $P(A \cap B) = 0.3$ (A1)

[6 marks]

7. Given that $\frac{dy}{dx} = \cos\left(x - \frac{\pi}{4}\right)$ and $y = 2$ when $x = \frac{3\pi}{4}$.

Recognizing integration is needed: (M1)

$y = \sin\left(x - \frac{\pi}{4}\right) + c$ (A1)

Substituting $x = \frac{3\pi}{4}$, $y = 2$: (M1)

$2 = \sin\left(\frac{3\pi}{4} - \frac{\pi}{4}\right) + c = \sin \frac{\pi}{2} + c = 1 + c$

$c = 1$

$y = \sin\left(x - \frac{\pi}{4}\right) + 1$ (A1)

[4 marks]

8. Consider $f(x) = -(x - h)^2 + 2k$ and $g(x) = e^{x-2} + k$.

(a) $f'(x) = -2(x - h)$ (A1)

(b) $g'(x) = e^{x-2}$, so $g'(3) = e$ (A1)

Common tangent at $x = 3$ means $f'(3) = g'(3)$: (M1)

$$-2(3 - h) = e$$

$$3 - h = -\frac{e}{2}$$

$$h = 3 + \frac{e}{2} = \frac{e + 6}{2}$$
 (A1)

(c) Common tangent means $f(3) = g(3)$: (M1)

$$-(3 - h)^2 + 2k = e^{3-2} + k$$

$$-\left(-\frac{e}{2}\right)^2 + 2k = e + k$$
 (A1)

$$-\frac{e^2}{4} + 2k = e + k$$

$$k = e + \frac{e^2}{4}$$
 (A1)

[7 marks]

9. $A = \int_0^c \frac{x}{x^2 + 2} dx$

Using substitution $u = x^2 + 2$, $du = 2x dx$: (M1)

$$= \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln |u| = \frac{1}{2} \ln(x^2 + 2)$$
 (A1)

$$\left[\frac{1}{2} \ln(x^2 + 2) \right]_0^c = \frac{1}{2} \ln(c^2 + 2) - \frac{1}{2} \ln 2$$
 (M1)

Setting equal to $\ln 3$:

$$\frac{1}{2} \ln\left(\frac{c^2 + 2}{2}\right) = \ln 3$$
 (M1)

$$\ln\left(\frac{c^2 + 2}{2}\right) = \ln 9$$

$$\frac{c^2 + 2}{2} = 9$$
 (A1)

$$c^2 = 16$$

$$c = 4$$

(A1)

(Since $c > 0$, reject $c = -4$.)

[6 marks]

Total: 52 marks